

Project Proposal: Disentangling Predictive Surprisal, Entropy, and Syntactic Integration Cost in Human Sentence Processing

Harshit Gupta
2022101124

Sanchit Jalan
2022101070

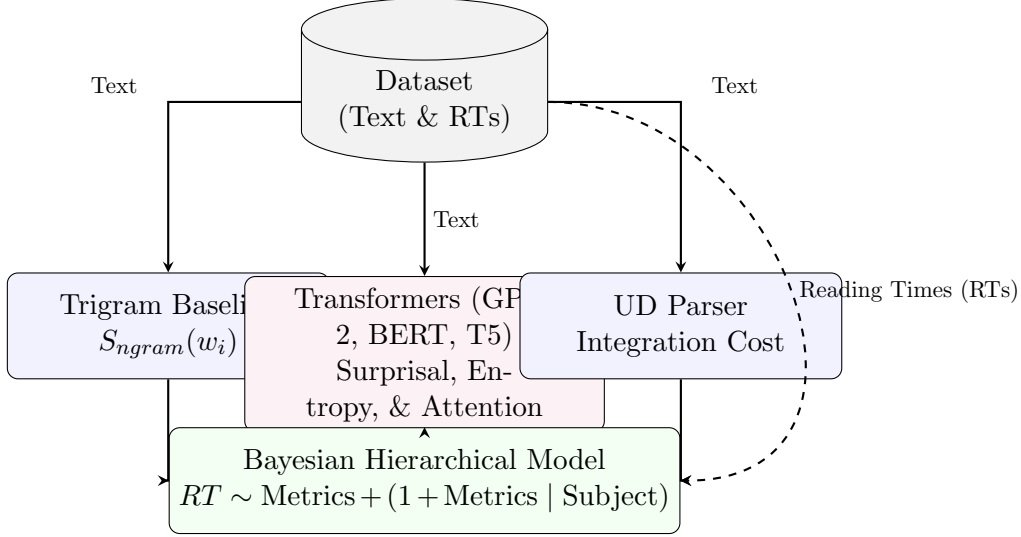
Research Questions & Hypotheses

This project investigates the cognitive mechanisms underlying human reading difficulty by evaluating Information-Theoretic metrics (Surprisal and Entropy) against Memory-Based Integration Cost across different neural architectures, utilizing Bayesian hierarchical modeling.

- **Hypothesis 1 (Deep vs. Shallow Prediction):** Deep neural surprisal (derived from Transformer models) will predict human reading times significantly better than local n-gram surprisal, indicating that human sentence processing relies on long-distance contextual cues rather than just local collocations.
- **Hypothesis 2 (Prediction vs. Memory):** Syntactic integration cost (dependency length) will no longer explain a statistically significant portion of reading time variance once deep neural surprisal is fully accounted for within the Bayesian posterior distribution.
- **Hypothesis 3 (Surprisal vs. Entropy):** Both the unexpectedness of a specific word (Surprisal) and the model's underlying uncertainty about the upcoming context (Entropy) will provide distinct, independent predictive power for human reading times.
- **Hypothesis 4 (Architectural Modality):** Autoregressive models (GPT-2) will yield superior fits to reading behavior compared to bidirectional (BERT) or encoder-decoder (T5) architectures, as strictly left-to-right processing mirrors human sequential reading.
- **Hypothesis 5 (Mechanistic Syntactic Tracking):** Specific attention heads within the Transformer models will naturally correlate with dependency length (Integration Cost), suggesting that these models implicitly learn structural working memory constraints.
- **Hypothesis 6 (Individual Reader Strategies):** Bayesian hierarchical modeling will reveal significant per-reader variance (random slopes), demonstrating that certain readers rely more heavily on predictive processing while others are more sensitive to structural memory load.

Methods

The project will follow a computational psycholinguistics pipeline, aligning behavioral data with computational metrics to isolate the drivers of reading difficulty.



1. **Behavioral Data Preparation:** We will extract word-by-word reading times to serve as our baseline dependent variable. Each word’s reading time will be logged in milliseconds.
2. **Extracting N-Gram Surprisal (Baseline Predictability):** We will train a trigram model with Kneser-Ney smoothing on a large background corpus to generate a local surprisal value for each word: $S_{ngram}(w_i) = -\log_2 P(w_i \mid w_{i-2}, w_{i-1})$.
3. **Extracting Contextual Entropy & Neural Surprisal:** Using HuggingFace, we will extract metrics from GPT-2, BERT (using pseudo-surprisal via masking), and T5. For each word, we will calculate Neural Surprisal ($S = -\log P(w_i \mid \text{context})$) and Contextual Entropy ($H = -\sum P(x) \log P(x)$) over the vocabulary distribution.
4. **Extracting Attention Weights:** We will extract the self-attention matrices ($A^{(l,h)}$) from the Transformer models to identify specific heads that align with syntactic dependency structures.
5. **Extracting Integration Cost (Memory Load):** Utilizing Universal Dependencies (UD) parses, we will calculate the linear distance between each word and its syntactic head to quantify the working memory load required to integrate the word into the sentence structure.
6. **Bayesian Hierarchical Modeling:** We will implement Bayesian hierarchical models (e.g., using `brms` in R or `PyMC` in Python) to estimate full posterior distributions. We will model random slopes for Surprisal and Integration Cost per subject, allowing us to quantify individual reader strategies and test the explanatory power of our metrics.

Dataset(s)

We will use one of the following high-quality psycholinguistic datasets for this project, all of which provide strictly aligned word-by-word reading metrics:

- **Natural Stories Corpus:** ~10,000 words of English narrative text with word-by-word self-paced reading times and annotated UD parses. (<https://github.com/languageMIT/naturalstories>)

- **GECO (Ghent Eye-Tracking Corpus):** A complete English novel (*The Mysterious Affair at Styles*, ~54,000 words) with eye-tracking data, offering highly rich, long-distance contextual cues. (<https://expsy.ugent.be/downloads/geco/>)
- **Dundee Corpus:** 20 complete newspaper articles (~51,000 words) with eye-tracking data. Widely considered the industry standard for testing integration cost against surprisal. *Note: The raw dataset is not freely available for direct public download due to newspaper copyright licensing. Researchers must request access directly from the original authors or via institutional linguistic data consortiums.*

Note: A background training dataset (such as a subset of Wikipedia or the BookCorpus) will also be used to train the baseline n -gram model.

Key Research Paper References

1. Hale, J. (2001). *A probabilistic Earley parser as a psycholinguistic model*. In Proceedings of NAACL.
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3. Demberg, V., & Keller, F. (2008). *Data from eye-tracking corpora as evidence for theories of syntactic processing complexity*. Cognition, 109(2), 193-210.
4. Oh, B., & Schuler, W. (2023). *Transformer-based Language Models Explain Human Reading Times*. In Proceedings of CogSci.
5. Wilcox, E. G., Gauthier, J., Hu, J., Qian, P., & Levy, R. (2020). *On the Predictive Power of Neural Language Models for Human Real-Time Comprehension Behavior*. Proceedings of CogSci.
6. Clark, K., Khandelwal, U., Levy, O., & Manning, C. D. (2019). *What Does BERT Look At? An Analysis of BERT's Attention*. Proceedings of BlackboxNLP. (Added for attention head analysis).